TIC 378858431

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

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BLS EPHEMERIS
  Period       9.422447 d  (ALIASED, see below)
    -> corrected period: 18.844894 d
  Epoch (t0)   2891.0851 BTJD
  Duration     0.0840 d
  Depth        0.857 %  (8,570 ppm)
  SNR          291.8

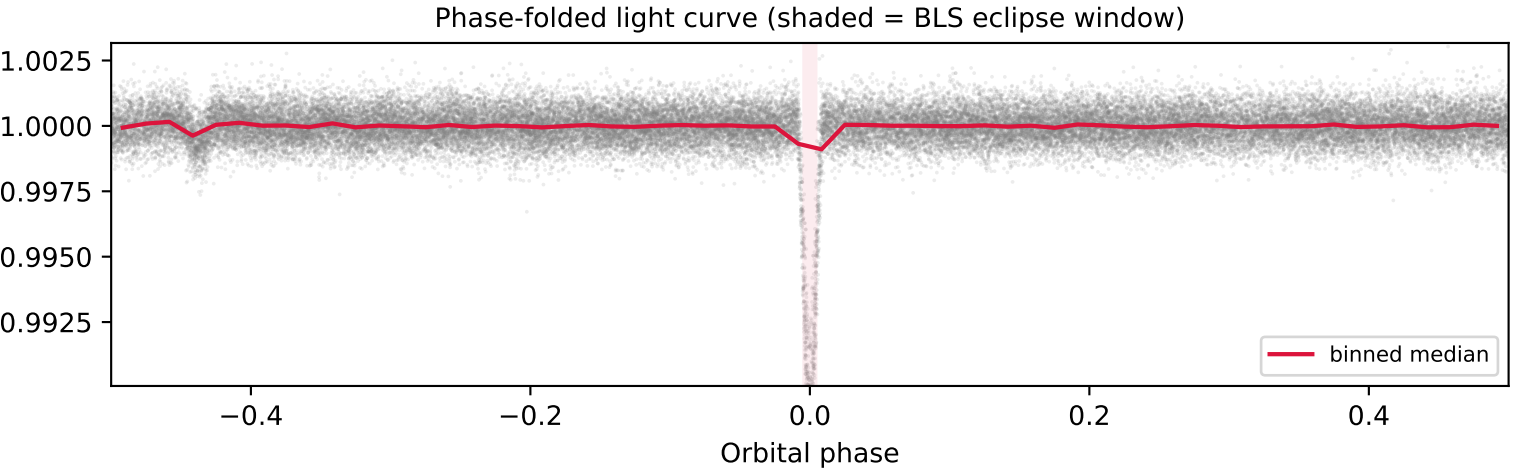
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     13  (gate >= 3)
  Robust depth ratio    1.03  (gate >= 0.50)
  Robust depth sigma    233.2  (gate >= 7.0)
  Odd/even z-score      4.24  (ALIAS FLAGGED)
  Secondary eclipse     detected, phase -0.43, 6.3 sigma
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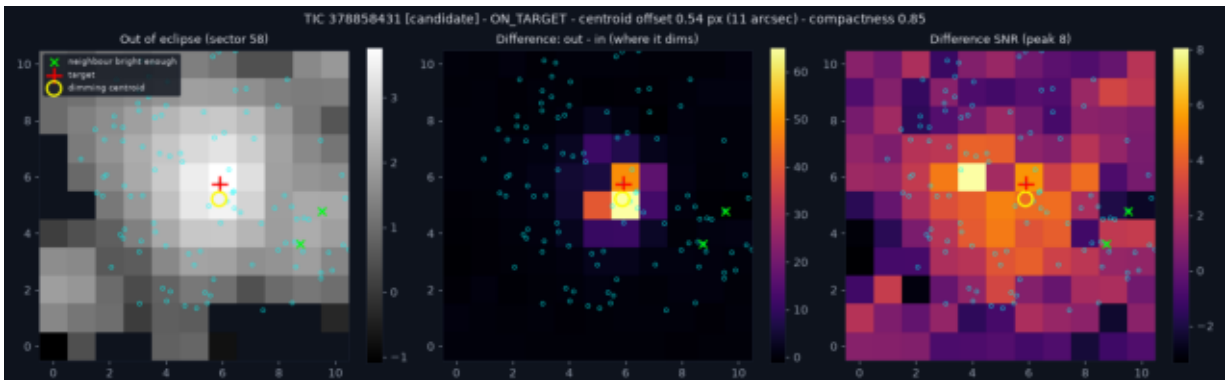
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 58    light-curve dip in this sector: 117.2 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   8.09   (gate >= 5.0)      Compactness   0.85   (gate >= 0.50)
  Dimming centroid offset     0.54 px (11 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 378858439
    3.28 px away, dTmag +4.71 - farther from the dimming than the target
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Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 4.24 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 18.844894 d, in any write-up.